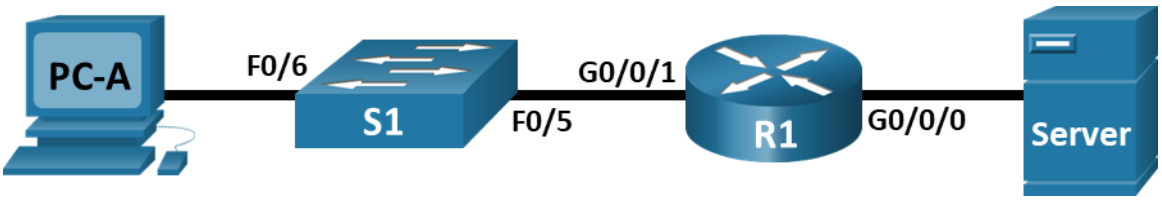
# Packet Tracer - Configure Basic Router Settings - Physical Mode

## Topology




## Addressing Table

| Device | Interface | IP Address / Prefix | Default Gateway |
|---|---|---|---|
| R1 | G0/0/0 | 192.168.0.1 /24 | N/A |
| | | 2001:db8:acad::1 /64 | |
| | | fe80::1 | |
| | G0/0/1 | 192.168.1.1 /24 | |
| | | 2001:db8:acad:1::1 /64 | |
| | | fe80::1 | |
| | Loopback0 | 10.0.0.1 /24 | |
| | | 2001:db8:acad:2::1 /64 | |
| | | fe80::1 | |
| PC-A | NIC | 192.168.1.10 /24 | 192.168.1.1 |
| | | 2001:db8:acad:1::10 /64 | fe80::1 |
| Server | NIC | 192.168.0.10 /24 | 192.168.0.1 |
| | | 2001:db8:acad::10 /64 | fe80::1 |

## Objectives

**Part 1: Set Up the Topology and Initialize Devices**

**Part 2: Configure Devices and Verify Connectivity**

**Part 3: Display Router Information**

## Background / Scenario

This is a comprehensive Packet Tracer Physical Mode (PTPM) activity to review previously covered IOS router commands. In Parts 1 and 2, you will cable the equipment and complete basic configurations and interface settings on the router.

In Part 3, you will use SSH to connect to the router remotely and use the IOS commands to retrieve information from the device to answer questions about the router.

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For review purposes, this activity provides the commands necessary for specific router configurations.

## Instructions

### Part 1: Set Up the Topology and Initialize Devices

#### Step 1: Cable the network as shown in the topology.

a. Click and drag the **Cisco 4321 ISR**, the **Cisco 2960 Switch**, and the **Server** from the **Shelf** to the **Rack**.

b. Click and drag the **PC** from the **Shelf** to the **Table**.

c. Cable the devices as specified in the topology diagram. Use **Copper Straight-through** cables for network connections.

d. From the **PC**, connect a **Console Cable** to the **Cisco 4321 ISR**.

e. Power on the **Cisco 4321 ISR**, **PC-A**, and **Server**. The power button for **Server** is on the bottom right. The 2960 switch should power on automatically.

### Part 2: Configure Devices and Verify Connectivity

#### Step 1: Configure the PC interfaces.

a. Configure the IP address, subnet mask, and default gateway settings on **PC-A**.

b. Configure the IP address, subnet mask, and default gateway settings on **Server**.

#### Step 2: Configure the router.

a. Console into the router and enable privileged EXEC mode.

b. Enter configuration mode.

c. Assign a device name to the router.

d. Set the router's domain name as ccna-lab.com. ip domain-name ccna-lab.com

e. Encrypt the plaintext passwords. service password-encryption

f. Configure the system to require a minimum 12-character password. security passwords min-length 12

g. Configure the username **SSHadmin** with an encrypted password of **55Hadm!n2020**. username SSHadmin secret 55Hadm!n2020

h. Generate a set of crypto keys with a 1024 bit modulus. crypto key generate rsa 1024

i. Assign **$cisco!PRIV*** as the privileged EXEC password. enable secret $cisco!PRIV*

j. Assign **$cisco!!CON*** as the console password. Configure sessions to disconnect after four minutes of inactivity, and enable login. line console 0 password $cisco!!CON* login

k. Assign **$cisco!!VTY*** as the vty password. Configure the vty lines to accept SSH connections only. Configure sessions to disconnect after four minutes of inactivity, and enable login using the local database. line vty 0 15 password $cisco!!VTY* login transport input ssh

l. Create a banner that warns anyone accessing the device that unauthorized access is prohibited. banner motd ##

m. Enable IPv6 routing. ipv6 unicast-routing

n. Configure all three interfaces on the router with the IPv4 and IPv6 addressing information from the addressing table above. Configure all three interfaces with descriptions. Activate all three interfaces.

The router should not allow vty logins for two minutes if three failed login attempts occur within 60 seconds.

o. Set the clock on the router.

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p. Save the running configuration to the startup configuration file.

What would be the result of reloading the router prior to completing the **copy running-config startup-config** command? Everything would be lost as it was configured from new.

### Step 3: Verify network connectivity.

a. Using the command line at **PC-A**, ping the IPv4 and IPv6 addresses for **Server**.

Were the pings successful? Yes

b. From **PC-A**, remotely access **R1** using the Telnet / SSH client.

Using the Telnet / SSH client on PC-A, open an SSH session to the R1 Loopback interface IPv4 address. Ensure that the Connection Type is set to **SSH** and use **SSHadmin** as the username. When prompted, enter the password **55Hadm!n2020**.

Was remote access successful?

c. Using the Telnet / SSH client on **PC-A**, open an SSH session to the R1 Loopback interface IPv6 address. Ensure that the Connection Type is set to **SSH** and use **SSHadmin** as the username. When prompted, enter the password **55Hadm!n2020**.

Was remote access successful? Yes

Why is the Telnet protocol considered to be a security risk?

Unsecure, plaintext credentials exposed

## Part 3: Display Router Information

In Part 3, you will use **show** commands from an SSH session to retrieve information from the router.

### Step 1: Establish an SSH session to R1.

Using Telnet / SSH client on PC-A, open an SSH session to the R1 Loopback interface IPv6 address and log in as **SSHadmin** with the password **55Hadm!n2020**.

### Step 2: Retrieve important hardware and software information.

a. Use the **show version** command to answer questions about the router.

What is the name of the IOS image that the router is running?

03.16.05.S -> 15.5

How much non-volatile random-access memory (NVRAM) does the router have?

32768K bytes of non-volatile configuration memory.

How much Flash memory does the router have?

3223551K bytes of flash memory at bootflash:.

b. The **show** commands often provide multiple screens of outputs. Filtering the output lets you display certain sections of the output. To enable the filtering command, enter a pipe (**|**) character after a **show** command, followed by a filtering parameter and a filtering expression. You can match the output to the

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filtering statement by using the **include** keyword to display all lines from the output that contain the filtering expression. Filter the **show version** command, using **show version | include register** to answer the following question.

What would be the boot process for the router on the next reload if the configuration register was 0x2142?

Configuration register 0x2142 tells the router to ignore its saved startup configuration during the next boot. Used for password recovery.

### Step 3: Display the startup configuration.

a. Use the **show startup-config** command on the router to answer the following question.

How are passwords presented in the output?

4

b. Use the **show running-config | section vty** command.

What is the result of using this command?

Displays line vty 0 4 and 5 15 their details and allowing SSH communication.

### Step 4: Display the routing table on the router.

Use the **show ip route** command on the router to answer the following questions.

What code is used in the routing table to indicate a directly connected network?

C

How many route entries are coded with a C code in the routing table?

3

### Step 5: Display a summary list of the interfaces on the router.

a. Use the **show ip interface brief** command on the router to answer the following question.

What command changed the status of the Gigabit Ethernet ports from administratively down to up?

no shutdown

b. Use the **show ipv6 int brief** command to verify IPv6 settings on R1.

What is the meaning of the [up/up] part of the output?

port up & protocol up

c. On **Server**, change its configuration so that it no longer has a static IPv6 address. Then, issue the **ipconfig** command on **Server** to examine the IPv6 configuration.

What is the IPv6 address assigned to Server?

2001:DB8:ACAD:0:207:ECFF:FEA5:6286

What is the default gateway assigned to **Server**?

FE80::1

From **PC-B**, issue a ping to the **R1** default gateway link local address. Was it successful?

Yes

From **Server**, issue a ping to the **R1** IPv6 unicast address 2001:db8:acad::1. Was it successful?

Yes

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## Reflection Questions

1. In researching a network connectivity issue, a technician suspects that an interface was not enabled. What **show** command could the technician use to troubleshoot this issue?

2. In researching a network connectivity issue, a technician suspects that an interface was assigned an incorrect subnet mask. What **show** command could the technician use to troubleshoot this issue?

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